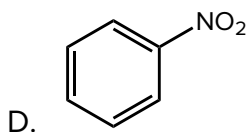
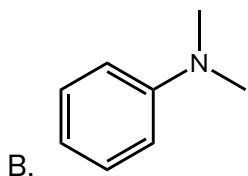
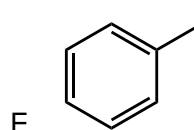
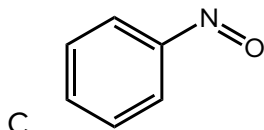
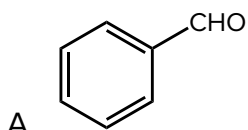


I. Multiple-Choice Questions

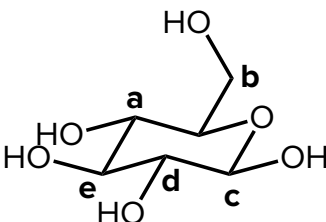
Choose the best answer among the choices provided. Circle your choice *and* write its letter in the box next to each question.

1. Treatment of diphenylmethanol with strong acid generates the diphenylmethyl cation, a good electrophile. Which of the following compounds reacts most rapidly with this intermediate?

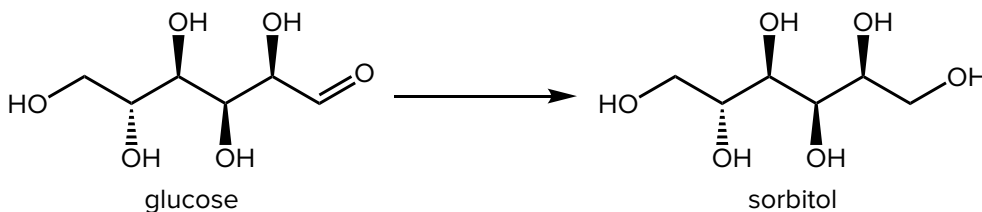


2. Which of the labeled carbons undergoes nucleophilic substitution when the monosaccharide below is treated with acid and a nucleophile?

- A. Carbon a
B. Carbon b
C. Carbon c
D. Carbon d
E. Carbon e



3. Sorbitol is a sugar alcohol used as a sucrose substitute that can be synthesized in a single step from glucose. What type of reaction converts glucose into sorbitol?



- A. Elimination
B. Nucleophilic substitution
C. Electrophilic substitution
D. Oxidation
E. Reduction

4. Which of the following amino acids has a side chain that is incapable of hydrogen bonding? Consult the page of amino acid structures at the end of the exam.

☐

- A. Tryptophan
- B. Tyrosine
- C. Serine
- D. Alanine
- E. Asparagine

5. Which of the following amino acids is most likely to be found on the exterior of a folded protein? Consult the page of amino acid structures at the end of the exam.

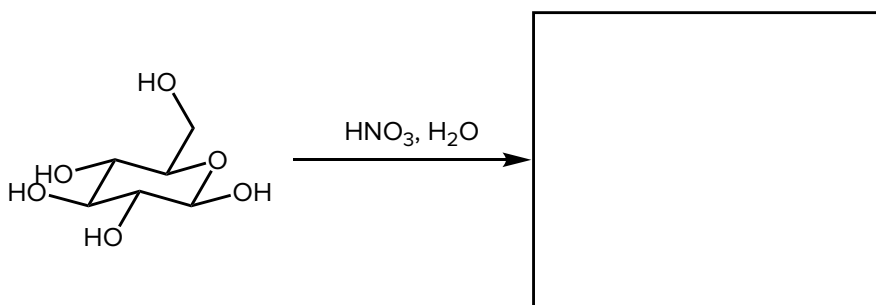
☐

- A. Aspartic acid (aspartate)
- B. Valine
- C. Isoleucine
- D. Glycine
- E. Phenylalanine

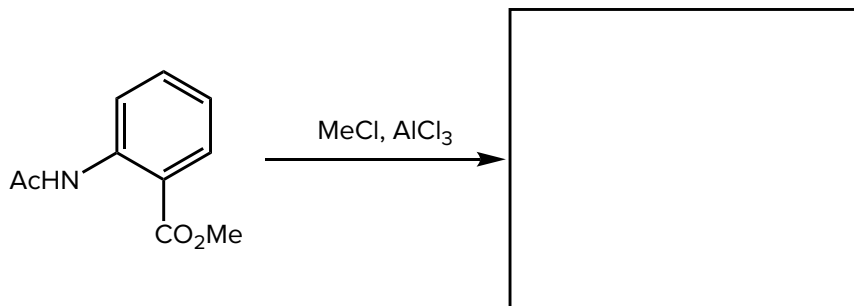
II. Predict the Products

Provide the major neutral organic product for each reaction or sequence of reactions below after appropriate workup. Clearly indicate the configurations of all stereocenters (except for those with random configurations; stereochemistry can be omitted at these).

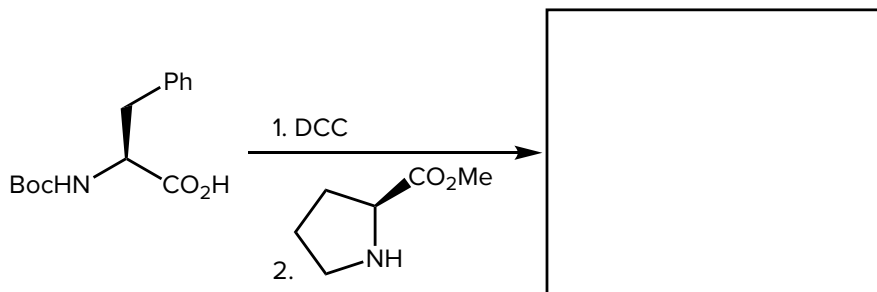
1.



2.



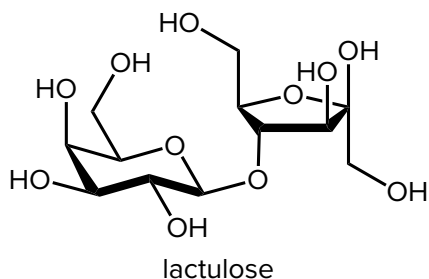
3.



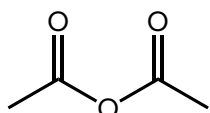
III. Short-answer Questions

For each question below, provide the requested structure, mechanism, or explanation. Ensure that your complete response fits within the box provided.

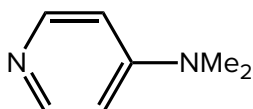
1. The structure of the disaccharide lactulose is shown below. The left-hand monosaccharide is galactose and the right-hand monosaccharide is fructose.



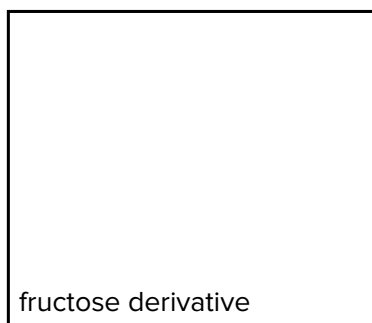
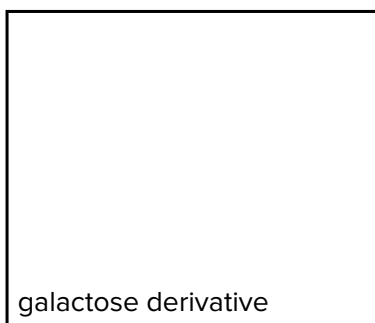
Draw both products formed when lactulose is treated with (1) an excess of acetic anhydride (Ac_2O) and DMAP followed by (2) aqueous sulfuric acid.



acetic anhydride

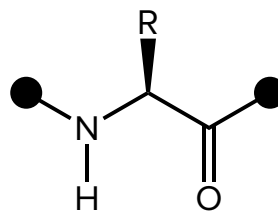


DMAP



2. A hypothetical amino acid residue appearing within a polypeptide chain is provided at right. Circle the two bonds in the polypeptide backbone that are capable of free rotation in this residue.

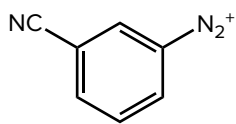
Note: ignore the side chain.



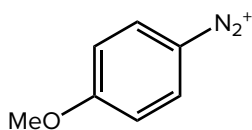
3. Rank the diazonium salts below from most reactive to least reactive with H_3PO_2 , a reducing agent that replaces the $-\text{N}_2^+$ group with $-\text{H}$.



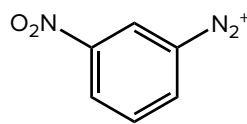
1



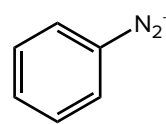
2



3



4



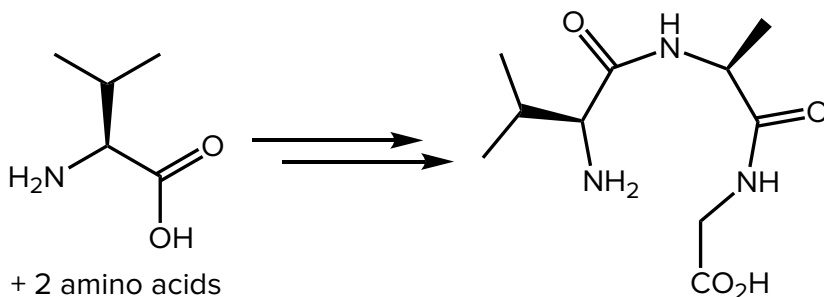
5

least reactive

most reactive

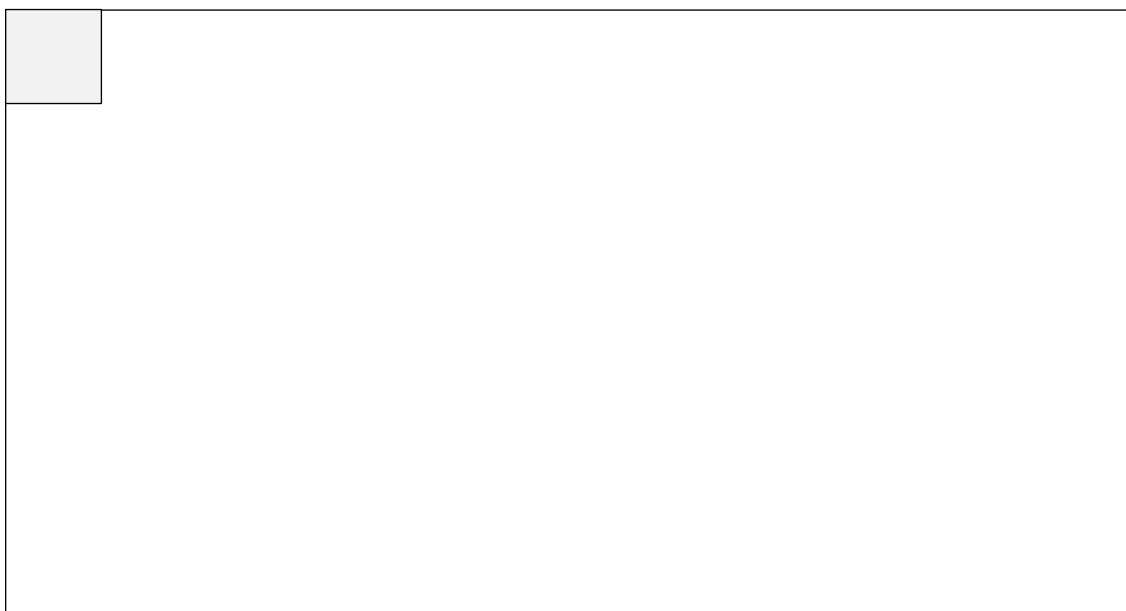
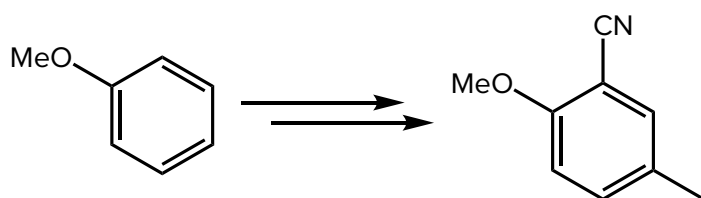
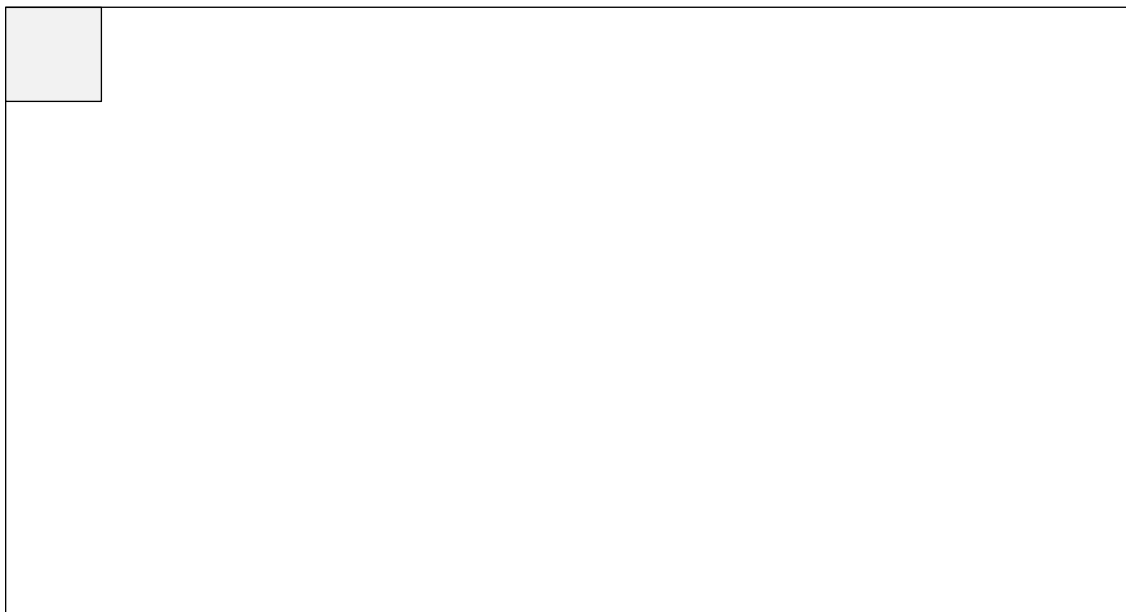
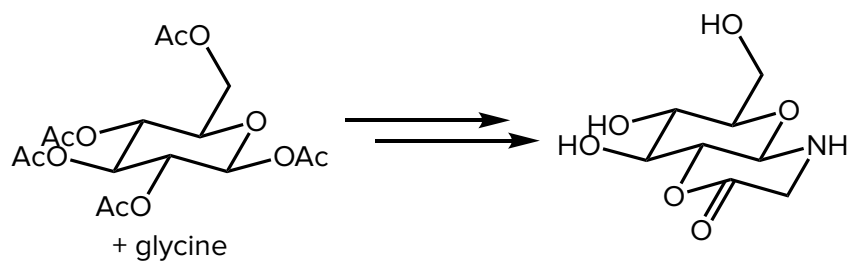
IV. Synthetic Planning

Of the three problems below, **choose two** and provide a sequence of reaction conditions that will accomplish the desired transformation. Include all isolated compounds, but do not draw curved arrows or any reactive intermediates. **Place an X in the shaded box next to the two problems you would like graded.**



1. + 2 amino acids

☐



V. Reaction Mechanism

To complete the reaction mechanism below, add curved arrows to the reactants of each elementary step and draw the products in the boxes provided. Each box should include a set of reactants and curved arrows except for the last, which will only include products. **NOTE:** None of the other possible anomer forms. Your mechanism should account for this!

